2018 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A skier of mass 62 kg is pulled up a slope by a motor-driven cable at a **constant speed** of $2.0\ \mathrm{m\ s^{-1}}$.
The slope is inclined at an angle of 17° to the horizontal.
The **frictional force** acting on the skier is $80\ \mathrm{N}$.

(a) Calculate the component of the skier's weight acting down the slope.

$$W = mg = 62 \times 9.8 = 607.6\ \mathrm{N}$$

Component **down the slope**:

$$W_{\text{parallel}} = mg\sin\theta = 607.6 \times \sin(17^\circ) = 177.8\ \mathrm{N}$$

(b) Calculate the force exerted by the cable on the skier.

Since the skier is moving at **constant speed**, net force = 0.
So, cable must balance both:

- Component of weight down slope: 177.8 N
- Friction: 80 N

$$F_{\text{cable}} = 177.8 + 80 = 257.8\ \mathrm{N}$$

(c) Calculate the work done by the cable in pulling the skier 15 m up the slope.

$$W = Fd = 257.8 \times 15 = 3867\ \mathrm{J}$$

(d) Calculate the power output of the cable.

$$P = \frac{W}{t}$$

First, calculate time:

$$t = \frac{d}{v} = \frac{15}{2.0} = 7.5\ \mathrm{s}$$

$$P = \frac{3867}{7.5} = 515.6\ \mathrm{W}$$

🧠 Revision Tips

- Always resolve weight **down a slope** using $W\sin\theta$
- Constant speed = **balanced forces**, so cable force = resistive + weight component
- Work done: $W = Fd$, only the component **in the direction of motion** counts
- Power is **energy per second**: $P = \frac{E}{t}$